

Philosophical Foundations — Module 06: Learning

— Study Questions

1. How does Kuhn's cycle of normal science → crisis → revolution map onto parameter learning → model selection?
2. How does Lakatos's hard core / protective belt distinction relate to hierarchical model structure?
3. How does Popper's falsification relate to Bayesian model evidence decline?
4. Explain how free energy $F = \text{Complexity} - \text{Accuracy}$ formalizes Occam's Razor.
5. What is Bayesian Model Reduction? How does it implement formal parsimony?
6. What is the demarcation problem? How does Active Inference address it?
7. How can pseudoscience be characterized in Active Inference terms?
8. What is dogmatic belief updating? How do overly strong priors prevent learning?
9. State Hume's problem of induction. How does Active Inference respond?
10. How does evolutionary history shape inductive priors?
11. What is under-determination? How does model comparison address it?
12. Compare scientific realism with constructive empiricism. Where does the FEP stand?
13. How does model evidence relate to empirical adequacy?
14. Can Active Inference model scientific communities as multi-agent belief-updating systems?
15. How does precision relate to the weight of evidence in theory change?
16. What is the role of surprise in driving scientific discovery?
17. How does the FEP's treatment of learning differ from classic empiricist accounts?
18. Can the FEP explain irrational belief persistence (conspiracy theories, ideological rigidity)?
19. How does Bayesian model comparison differ from classical hypothesis testing?
20. Critically evaluate: Does the FEP offer genuine insight into the nature of science, or does it merely provide a mathematical metaphor?